



# MANIPAL ACADEMY OF HIGHER EDUCATION

**Manipal Institute of Technology, Bengaluru**

BTech VI Semester

End Semester Examination - April / May -2025

**EMBEDDED SYSTEM DESIGN [ECE-4411]**

**Marks: 50**

**Duration: 180 mins.**

## PART-A

**Answer all the questions.**

- 1) An embedded-system designer must balance multiple design metrics while ensuring the required functionality of the system. Analyze how the optimization of one design metric might impact others, using suitable examples. (5)
  - A)
  - B) Explain the following IC technologies used to implement a processor. Elaborate the parameters of performance, design and cost (5)
    - i. Full custom/VLSI
    - ii. Semi custom ASIC
    - iii. PLD
- 2) Compare the four common software architectures used in embedded systems by applying your understanding of their structure, functionality, and suitability for different applications. Provide relevant examples to support your comparison. (4)
  - A)
  - B) Three processes with process IDs P1, P2, P3 with estimated completion time 10, 5, 7 milliseconds and priorities 0, 3, 2 {0-highest priority, 3-lowest priority} respectively enters the ready queue together. Calculate the waiting time and Turn Around Time (TAT) for each process and the Average waiting time and Turn Around Time (Assuming there is no I/O waiting for the processes) in priority based scheduling algorithm. (3)
    - C) Using appropriate code examples, apply your understanding of task synchronization to explain common issues such as race conditions and deadlocks. (3)
- 3) With suitable diagram illustrate the following circuits essential for the proper functioning of the processor/controller and firmware execution. (4)
  - A) i. Watchdog Timer  
ii. Real Time Clock
  - B) Differentiate between the Data Flow Graph (DFG) model and the Control Data Flow Graph (CDFG) model. How are data and control operations represented in each model? Support your answer with a suitable example. (3)
  - C) Demonstrate your understanding of out-of-circuit programming by explaining its role in integrating hardware and software during embedded system development. Support your explanation with a relevant example and its sequence of operations. (3)
- 4) Apply your knowledge of pointers to write a C program that swaps two integers using a function. Explain how pointers are used in your solution to achieve the swap. (5)

- A)
  - B) Write a C program that uses a struct to store student records and sorts them by roll number using bubble sort. (5)
- 5) Explain the concept of System on Chip (SoC) and its significance in modern embedded systems. Provide examples of applications where SoCs are commonly used. (5)
- A)
  - B) Elaborate the 3 primary objectives of EDLC (5)

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